

PC-Based Fingerprint Scanner



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Computerised fingerprint scanners have been in use in police stations, high-security buildings, time-attendance systems and even PC keyboards for quite some time. Here we describe the construction and working of such a PC-based fingerprint security system. It scans your fingerprint and registers it in a PC to compare the captured fingerprints with it.

Fingerprint basics

Every person has a unique fingerprint pattern. The pattern of ridges and valleys on fingers makes it easier for the hands to grip things, in the same way as a rubber tread pattern helps a tyre grip the road. There are three basic fingerprint patterns: loop, whorl and arch. Other common fingerprint patterns include the tented arch, the plain arch and the central pocket loop.

Consequently, fingerprints are a unique marker for a person—even an identical twin. Two fingerprints may look the same at a glance but a trained investigator or an advanced piece of software can pick out the difference.

Fingerprint analysis done in crime investigation and security systems uses the same technique. A fingerprint scanner takes the place of a human analyst by collecting a print sample and comparing it to other samples on record. Once identified by a fingerprint security system, a person can be allowed access to a place or something. For example, a fingerprint image can be used in place of a typed password for a computer system, or it can replace swipe cards that are used to gain access to a building. We will see in the following sections how the scanner detects and recognises different complex patterns.

The types of fingerprint security systems include optical, silicon chip,



Fig. 1: Typical fingerprint image



Fig. 2: R305 fingerprint scanner module

ultrasound and thermal. We will discuss here only the optical scanner that has been used in this project.

R305 optical scanner

R305 fingerprint optical scanner or sensor module (Fig. 2) has an optical glass plate and controller unit on its base. AS601 processor is used in the controller unit. It is a member of Synchip's Cordis 5+ family that is targeted at fingerprint identification and digital certification, and can be used in a wide range of embedded applications. It features SEA/RSA accelerator engines, embedded non-volatile memory (Flash/OTP), fingerprint processing accelerator and its algorithm firmware.

TABLE I
Specifications of R305 Fingerprint Module

Module	Specifications
Power	DC 4.5V-6.0V
Working current	Typical: 100 mA Peak: 150 mA
Baud rate	(9600×N) bps, N=1-12 (default N=6)
Image acquiring time	<0.5s
Storage capacity	256
FAR	<0.001%
Average searching time	<1s (1:1000)
Working environment	Temp: -10°C to +40°C RH: 40%-85%

Cordis 5+ is a 32-bit RISC core that features 16-/32-bit ISA and a Harvard bus architecture. Important specifications of R305 module are listed in Table I.

Module memory. There is an image buffer and two 512-byte character file buffers within the RAM space of R305 module. The content of these buffers are lost when the power goes off. Users can read and write any of the buffers through computer program instructions. ImageBuffer1 is used for image storage. CharBuffer1 and CharBuffer2 character file buffers can be used to store both the character file and the image template file. The image format is 256×288 pixels.

Fingerprint library. R305 module has some space within Flash reserved for fingerprint template storage or fingerprint library. Note that the capacity of the library depends on the capacity of Flash memory of a particular module. Fingerprint template is stored in Flash in sequential order. You can access the library through template number. The serial number of the template in the library is 0, 1, 2, 3 ... N. The content is not lost when the power goes off.

Optical scanner. The heart of the optical scanner is a charge-coupled device (CCD)—a light sensor system used in digital cameras and camcorder